

Time and Space complexity

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$\hookrightarrow O(n) \rightarrow$

$\theta(n) \rightarrow$ Avg

$\omega(n) \rightarrow$ Best

// Swap two numbers.

$a = 5, b = 6;$

int temp = a

~~int~~ a = b;

b = a;

constant
time

$O(3) \approx O(1)$

for(int i=0; i < ⁿarr.length; i++) {
 Sort(arr[i]);

} $\hookrightarrow O(n)$

↪

for(int i=0; i < n; i++) {

 for(int j=0; j < n; j++) {

 Sort("*");

T.C

$O(n^2)$

```

    }
}

```

```

for (int i=0; i<n; i++) {

```

```

    for (int j=0; j<l; j++) {

```

```

        Sort (*);

```

```

    }

```

```

}

```

$O(n^2)$

* points to remember.

① Neglect the Constants.

② Neglect the lower value.

```

int n = 64; 128

```

```

while (n > 0) {

```

```

    Sort (n);

```

```

    n = n/2;

```

```

}

```

k times → (k < n)

n = 64

n = 32

n = 16

n = 8

n = 4

n = 2

n = 1

$\frac{n}{2^k}$

2^k

Constraint

$< [10 \dots 11]$

$< [10 \dots 15]$

< 400

$< 10^4$

$< 10^5$

$< 10^8 / 10^9$

$< 10^9$

~~~~~  
~~~~~

$n = 0$

Time Complexity

$O(n!)$ $\rightarrow$

$O(2^n)$ $\rightarrow$

$O(n^3)$

$O(n^2)$

$O(n \log n)$

$O(n)$

$O(\log n)$

$O(1)$